More Series Solutions...

ODE solutions are not always as "smooth" as a power series, and we need to modify our approach to "see" these singularities.

Euler's Equation

"singular" at $x=0$ ↓ ; numbers ↓

$$x^2u''(x) + a\,xu'(x) + b\,u(x) = 0$$

$$A\,u = 0$$

"The origin ($x=0$) is a singular point of this ODE"

$u(x) \sim x^r \rightarrow \quad (r(r-1) + ra + b)\,x^r = 0$

For $a-1 = \frac{1}{2}$, $b = -\frac{1}{2}$ $\Rightarrow$ $r = \frac{1}{2}, -1$



$$\Rightarrow u(x) = c_1 x^{-1} + c_2 x^{1/2}$$

Example: $x^2u''(x) + 4xu'(x) + (x^2+2)u(x) = 0$

↑ singular point at $x=0$

$u(x)$ = $x^r \sum a_n x^n$

$$\sum_{n=0}^{\infty} (\quad\quad) x^{n+r} = 0$$

$=0$

Since x^{n+r} are lin. indep. functions

Convenient to organize matching in table:

coeffs of	x^r	x^{r+1}	x^{r+2}	...	x^{r+m}
x^2u''	$r(r-1)a_0$	$(r+1)ra_1$	$(r+2)(r+1)a_2$		$(m+r)(m+r-1)a_m$
$4xu'$	$4ra_0$	$4(r+1)a_1$	$4(r+2)a_2$		$4(m+r)a_m$
x^2u	0	0	a_0		a_{m-2}
$2u$	$2a_0$	$2a_1$	$2a_2$		$2a_m$

$$\Rightarrow \quad [r(r-1) + 4r + 2]a_0 = 0$$

$$\Rightarrow \quad [(r+1)r + 4(r+1) + 2]a_1 = 0$$

$$\Rightarrow \quad (m+r)(m+r-1)a_m + 4(m+r)a_m + a_{m-2} + a_m = 0$$

depend on r!

$$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_N \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$L(r)\,\underline{a} = 0 \qquad (A - rI)\,\underline{a} = 0$$

$$[L - p(r)\,I]\,\underline{a} = 0$$

↑ quadratic polynomial

Q: Which singularities does this ODE allow?

$a_1 = 0$, $a_0 = \text{free}$ $\Rightarrow$ $\left[r(r-1) + 4r + 2\right] a_0 = 0$

$$(r^2 + 3r + 2) = (r+2)(r+1) = 0$$

$$r = -1 \quad \text{and} \quad r = -2$$

$n=2$, $r=-1$

$$(n+r)(n+r-1)a_n + 4(n+r)a_n + a_{n-2} + a_n = 0$$

$$(n-1)(n-2)a_n + 4(n-1)a_n + a_{n-2} + a_n = 0$$

$$\underbrace{\left((n-1)(n-2) + 4(n-1) + 1\right)}_{p(n)} a_n = -a_{n-2}$$

$$a_n = \begin{cases} -\dfrac{a_{n-2}}{p(n)} & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

$a_0 = \text{free}$

$$u(x) = a_0 x^{-1} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} x^{2m}$$

$$= a_0 x^{-2} \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)$$

$$= a_0 \frac{\sin x}{x^2}$$ $\Leftarrow$ singular at $x = 0$

Fuchs's Theorem

(*) $u'' + f(x)u' + g(x)u = 0$

Idea: $x^2u'' + axu' + bu = 0 \Rightarrow u'' + \frac{a}{x}u' + \frac{b}{x^2}u = 0$

Def: The origin ($x=0$) is a regular singular point if $xf(x)$ and $x^2g(x)$ have conv. power series (Taylor series) at $x=0$.

Thm (Fuchs's): If (*) has a regular singular point, then there are either

(i) two lin. indep. solutions of the form

(distinct roots $r_\pm$) $\quad u_\pm(x) = x^{r_\pm} \sum_{n=0}^{\infty} a_n^\pm x^n$

(ii) two lin. indep. solutions of the for

$u_1(x) = x^r \sum_{n=0}^{\infty} a_n x^n$ and $u_2(x) = c_1 u_1(x) \ln|x| + x^r \sum_{n=0}^{\infty} b_n x^n$

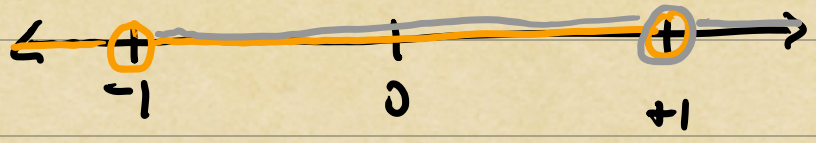
(repeated roots $r_+ = r_-$)

Q: What if the singular point is at $x_* \neq 0$?

$\Rightarrow$ Change variables $y = x - x_*$ (u, u', u'' don't change)

Q: What if there are multiple singular pts?

$\Rightarrow$ Different expansions around different singular points.



$u_{+1}(x) = x^{r_{+1}}(a_0 + a_1 x + \cdots$

$u_{-1}(x) = x^{r_{-1}}(b_0 + b_1 x + \cdots)$

Differential Eq's in Fourier Basis

(Taylor series)

In contrast to monomial basis, Fourier basis

$\Rightarrow$ global (pointwise) approximation of $C^1(-\pi, \pi)$

$\Rightarrow$ ONB for $L^2(-\pi, \pi)$

$\Rightarrow$ Fourier modes diagonalize differentiation (& integration)

$$\frac{d}{dx}\left[e^{ikx}\right] = ik\left[e^{ikx}\right]$$